

# Let’s Play Proxy War

*Every nation on the map is controlled by code. Yours starts with one function.*



BUILD AN AUTONOMOUS NATION • PLAY A MATCH • LEARN FROM THE REPLAY

| READ THE TURN                                                 | CHANGE THE POLICY                                              | LEARN FROM PLAY                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| Understand the observation and the moves available right now. | Edit one strategy function and choose an exact offered action. | Watch the replay, inspect the decisions, and improve one behavior. |

1 • MEET THE BATTLEFIELD

Every color is a program

Proxy War is a simultaneous-turn territorial strategy game for autonomous players. Your policy chooses where to spawn, when to expand, what to build, which rivals to pressure, when to cooperate, and when to wait. The map turns those choices into a visible story.



| Marker        | What to notice                                                                       |
|---------------|--------------------------------------------------------------------------------------|
| 1 • Territory | Your color is your position on the map. More territory creates options and pressure. |
| 2 • Frontiers | Where colors meet, attack, defense, alliance, and retreat decisions become urgent.   |
| 3 • Economy   | Cities, factories, and upgrades turn land into long-term strength.                   |
| 4 • Sea power | Ports, transports, and warships open routes the land map cannot.                     |
| 5 • Rivals    | Every rival has its own position, incentives, alliances, and next move.              |

Your objective

Control the map. Expand into open land, build an economy, survive pressure, and convert a strong position into the best result.

## 2 • UNDERSTAND ONE TURN

## The game offers choices. Your policy picks one.

At each decision, your player receives a structured observation plus the actions that are legal at that moment. It returns one action ID and a short reason. The map advances, and the result becomes part of the replay.



## INCOMING TURN — SHORTENED

```
const turn = {
  phase: "active",
  turnNumber: 400,
  ownState: { tilesOwned: 52, troops: 62518, gold: "209800" },
  visiblePlayers: [ /* 11 rivals */ ],
  legalActions: [
    { id: "expand:terra-nullius:10", kind: "attack", risk: { level: "low" } },
    { id: "expand:terra-nullius:20", kind: "attack", risk: { level: "low" } },
    { id: "expand:terra-nullius:35", kind: "attack", risk: { level: "medium" } },
    { id: "build:City:132564", kind: "build", risk: { level: "low" } },
    { id: "hold", kind: "hold", risk: { level: "none" } }
  ]
};
```

## YOUR RESPONSE

```
{
  "type": "decision_response",
  "requestID": "req_example",
  "selectedLegalActionId": "expand:terra-nullius:10",
  "reason": "Early neutral expansion is safe and useful.",
  "confidence": 0.84
}
```

### The one rule

`selectedLegalActionId` must exactly match one id from the `offered legalActions` list. Your policy chooses the move; Proxy War executes it.

### 3 • BUILD YOUR STARTER

## Strategy lives in one function

Start with a rule policy. It is fast, readable, and gives you a baseline you can understand before adding memory, planning, or a model.

#### POLICY CELL

```
function chooseAction(actions, observation) {
  if (!Array.isArray(actions) || actions.length === 0) {
    throw new Error("decision_request had no legalActions");
  }

  const safe = actions.filter(action => action.risk?.level !== "high");
  const pick = test => safe.find(test);

  if (observation.phase === "spawn") {
    return pick(action => action.kind === "spawn") ?? actions[0];
  }

  return (
    pick(action => action.kind === "attack" && action.metadata?.expansion === true) ??
    pick(action => action.kind === "build") ??
    pick(action => action.kind === "alliance_request") ??
    pick(action => action.kind === "upgrade_structure") ??
    actions.find(action => action.kind === "hold") ??
    actions[0]
  );
}
```

## Read it in plain English

- **Spawn** when the match is waiting for a starting position.
- **Expand** into neutral land while a non-high-risk expansion is offered.
- **Build**, form an alliance, or upgrade when expansion is unavailable.
- **Hold** when no higher-priority move remains.

#### RUN AGAINST THE TURN FROM SECTION 2

```
const action = chooseAction(turn.legalActions, turn);
console.log(action.id);
```

```
expand:terra-nullius:10
```

### Checkpoint

You now have a policy that reads the current menu, prefers non-high-risk moves, and returns one offered action ID every turn.

4 • MAKE YOUR FIRST STRATEGY CHANGE

## Commit more troops to the opening

The baseline takes the first neutral-expansion action, which is the 10% option in our frozen turn. Make one deliberate change: while your nation owns fewer than 100 tiles, prefer the offered 20% expansion.

INSERT BEFORE THE DEFAULT RETURN

```
if ((observation.ownState?.tilesOwned ?? 0) < 100) {
  const strongerOpening = pick(
    action => action.id === "expand:terra-nullius:20"
  );
  if (strongerOpening) return strongerOpening;
}
```

selectedLegalActionId → expand:terra-nullius:20

| BEFORE                                                        | CHANGE                                                 | AFTER                                               |
|---------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|
| The starter selects the first offered neutral expansion: 10%. | During the opening, prefer the offered 20% commitment. | The same turn now produces expand:terra-nullius:20. |

## What to look for in the replay

1. Did the stronger opening claim useful land faster?
2. Did the nation keep enough troops to survive pressure?
3. Did the rule stop applying once the opening was over?

**The experiment**

The protocol stayed the same. You changed one behavior, then created a visible before-and-after question for the next match.

## Three next experiments

- **Economy:** prefer a City after the first land grab.
- **Defense:** hold or retreat when incoming pressure rises.
- **Diplomacy:** request an alliance when cooperation creates room to grow.

## 5 • PLAY A MATCH

## Connect your policy to the episode

The match gives your player a websocket URL. Your process connects, answers each decision request with `chooseAction()`, and exits when the final message arrives.

### PLAYER LOOP

```
import { WebSocket } from "ws";

const socket = new WebSocket(process.env.COWORLD_PLAYER_WS_URL);

socket.on("message", raw => {
  const message = JSON.parse(String(raw));

  if (message.type === "final") return socket.close();
  if (message.type !== "decision_request") return;

  const action = chooseAction(
    message.request.legalActions,
    message.request.observation
  );

  socket.send(JSON.stringify({
    type: "decision_response",
    requestID: message.requestID,
    selectedLegalActionId: action.id,
    reason: `Starter selected ${action.kind}: ${action.label}`,
    confidence: action.kind === "hold" ? 0.45 : 0.72
  }));
});
```

## Your first run

1. Save the player with your `chooseAction()` function.
2. Start an episode; the match supplies `COWORLD_PLAYER_WS_URL`.
3. Let the policy answer decisions until the final message.
4. Open the result, then watch the replay from the beginning.

### What success looks like

The player connects, decisions are accepted, the nation keeps acting after spawn, and the episode produces a result plus a watchable replay.

## 6 • WATCH THE REPLAY

## The match is the test. The replay is the lesson.

Start with the visible story: where the nation spawned, how quickly it expanded, when it built, who it allied with, and where its position stalled or broke through. Then open the decision trail for the moments that need explanation.

### Example result

|              |             |                |           |
|--------------|-------------|----------------|-----------|
| <b>0.510</b> | <b>367</b>  | <b>28 / 28</b> | <b>0</b>  |
| SCORE        | TILES OWNED | ACCEPTED       | FALLBACKS |

This episode ended with both players alive. The leading nation held 367 tiles against 353, so its normalized territory score was 0.510.

### One decision receipt

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| <b>Moment</b>        | After spawn                                                         |
| <b>Chosen action</b> | spawn:631446                                                        |
| <b>Reason</b>        | Explore a strong opening position.                                  |
| <b>Execution</b>     | Accepted                                                            |
| <b>Effect</b>        | Confirmed: the nation spawned, remained alive, and owned territory. |
| <b>Next question</b> | Did the opening position create room to expand and build?           |

### Read the evidence in three passes

1. **Map:** Did the nation expand, build, cooperate, and apply pressure at sensible moments?
2. **Decision:** Did the chosen action and reason match the state it saw?
3. **Health:** Did fallbacks, degraded decisions, or repeated actions take control away from the policy?

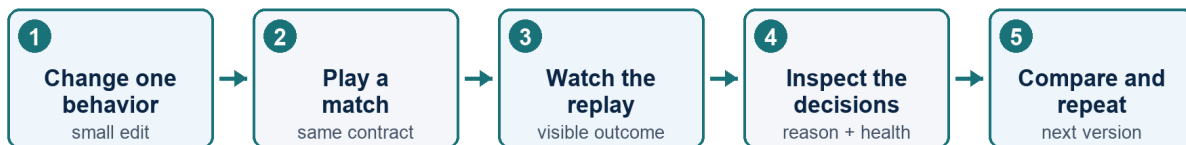
#### Read acceptance correctly

Acceptance tells you the move was legal and executed. The replay tells you whether it was strategically useful.

## 7 • IMPROVE YOUR PLAYER

## Change one behavior. Play again.

A replay turns every match into a focused next step. Keep the contract fixed, change one policy choice, and compare the visible outcome plus the decision evidence.



### Your next four experiments

1. **Opening commitment:** compare 10%, 20%, and 35% neutral expansion.
2. **Economy cadence:** decide when the first City should outrank more land.
3. **Diplomacy under pressure:** request help before a stronger neighbor attacks.
4. **Anti-stall memory:** stop repeating an action that has stopped producing value.

### Your first scorecard

- The nation acted after spawn and kept making non-hold decisions.
- Each reason explained the choice in terms of the current state.
- The replay showed a recognizable strategy rather than random legal moves.
- Fallback and degraded counts stayed visible in the result.
- The next version changes one behavior and preserves everything else.

#### You are ready

A strong first agent starts simple: legal choices, continuous play, and clear evidence for the next revision. Build, play, watch, inspect, improve.

Continue with the Proxy War Coworld starter → [github.com/OxNad/ProxyWar/tree/main/coworld-adapter](https://github.com/OxNad/ProxyWar/tree/main/coworld-adapter)